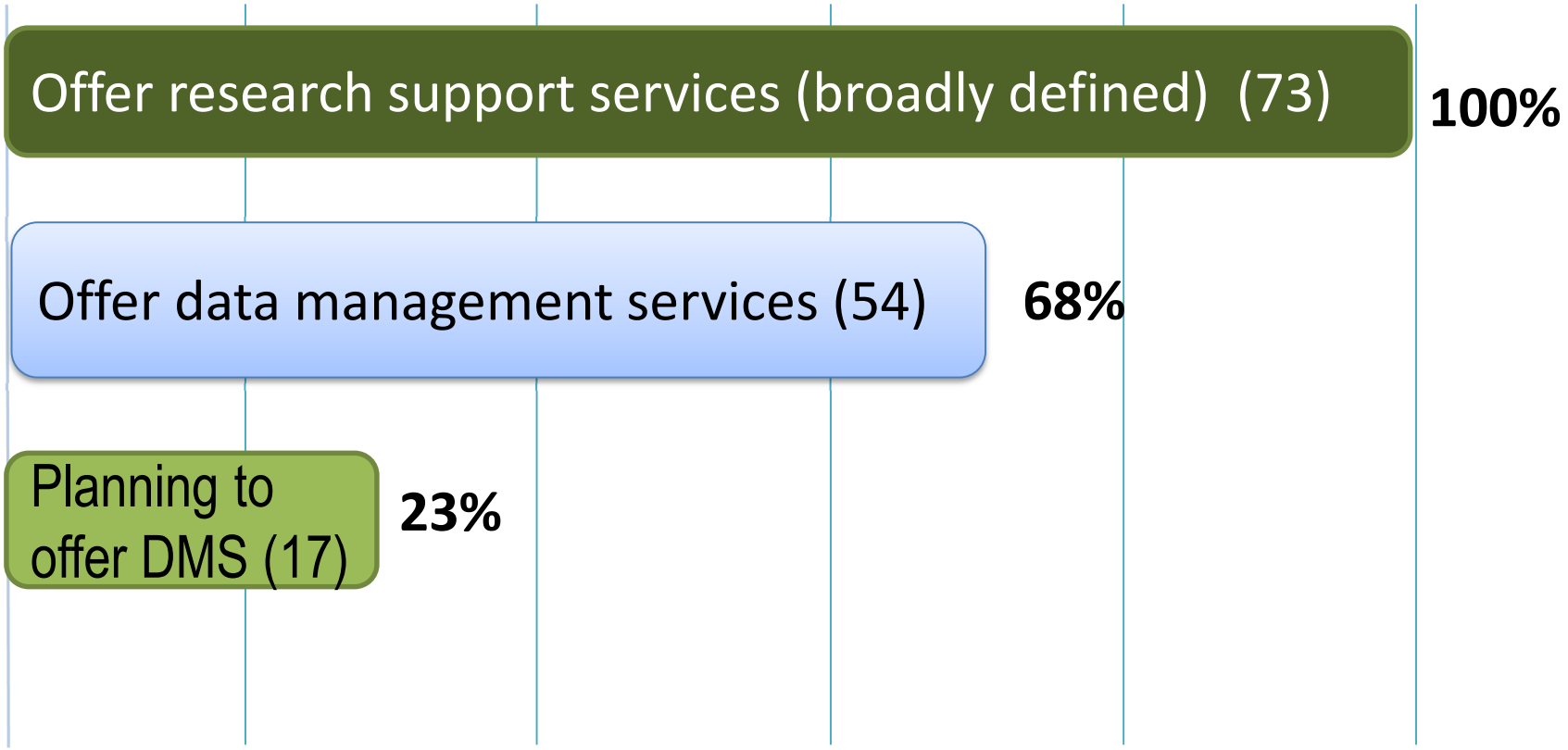
Streamlining the research data archival process at Johns Hopkins University

Dr. Jonathan Petters
Johns Hopkins Data Management Services
5 June 2014

Interest from Libraries in Providing Research Data Management Services (RDMS)



ARL SpecKit 334: Research Data Management Services, Fearon, David Jr.; Gunia, Betsy; Lake, Sherry; Pralle, Barbara E.; Sallans, Andrew L. July 2013.

Digital Collections Management is One Aspect of RDMS

Resources Required

http://commons.wikimedia.org/wiki/File:Video_archive.jpg

http://commons.wikimedia.org/wiki/File:Hazelton_coal_miners.jpg

A pilot project for the JHU Data Archive (Engineering research)

Size dependent thermodynamics and kinetics in electric field mediated colloidal crystal assembly†

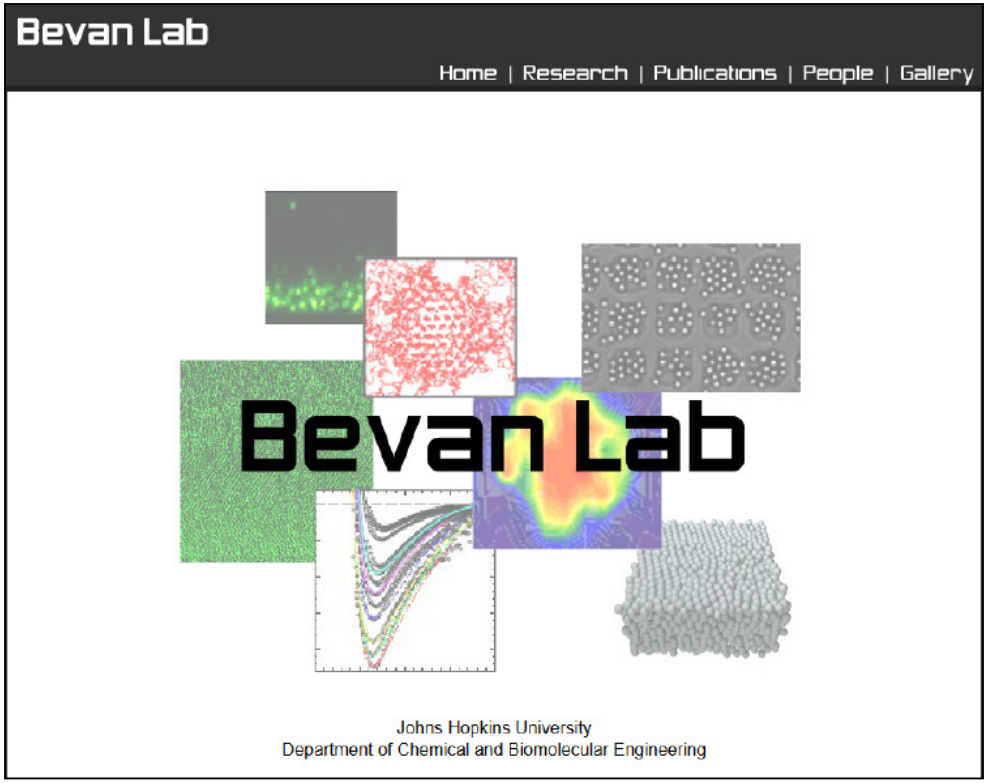
Archive is discipline and format agnostic

Size Dependent Thermodynamics and Kinetics in Electric Field Mediated Colloidal Crystal Assembly, Edwards, T.D.; Beltran-Villegas, D.J.; Bevan, M.A. *Soft Matter* Vol.**9**, 9208-9218, 2013

Pilots allow for experimentation!

Took several actions in preparing and depositing the data collection into the JHU Data Archive

~75 hours



http://www.jhu.edu/bevan/index.html

~23 Hours

Background Research

Size dependent thermodynamics and kinetics in electric field mediated colloidal crystal assembly†

Tara D. Edwards,‡ Daniel J. Beltran-Villegas‡ and Michael A. Bevan*

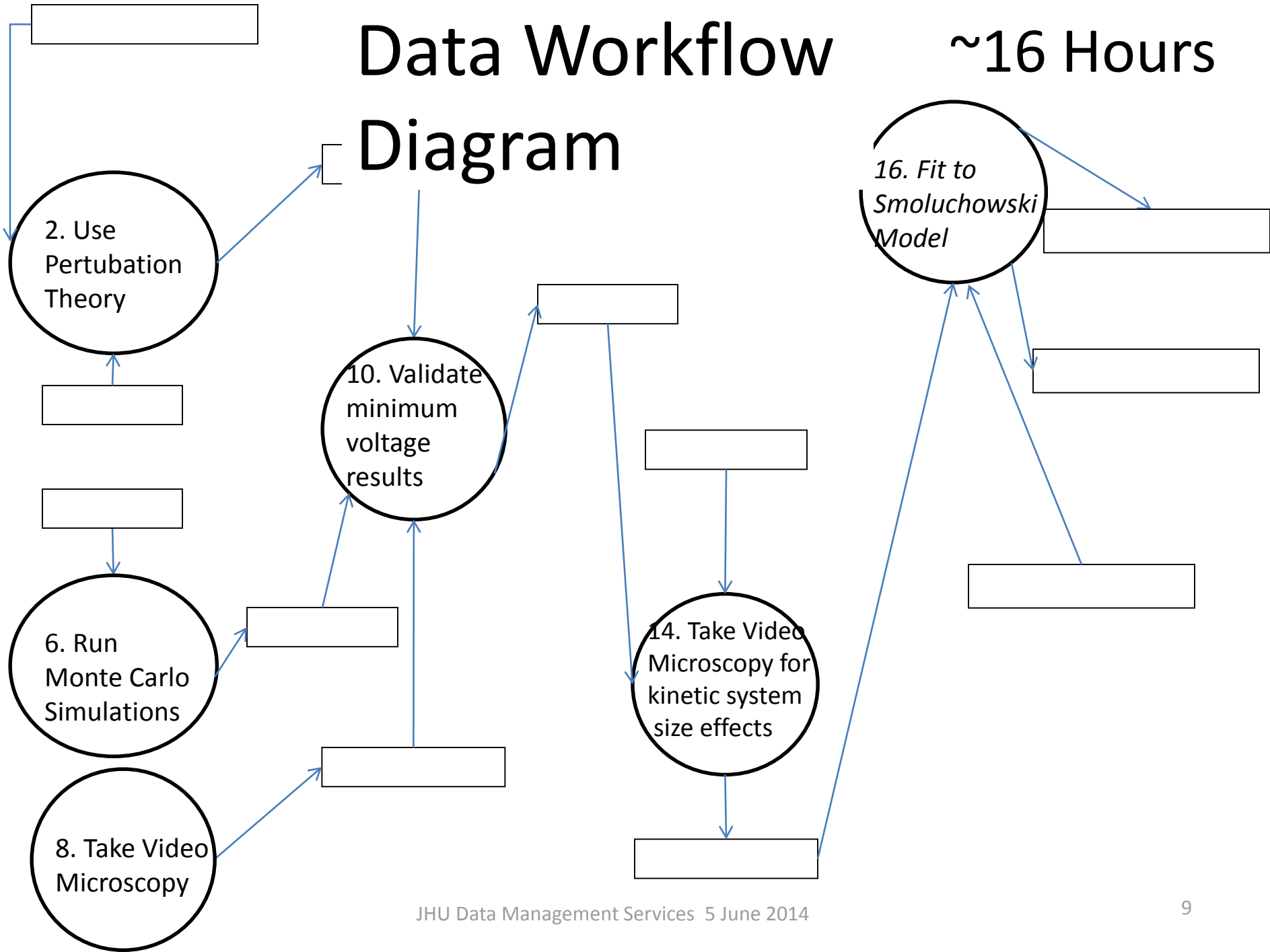
We compare measurements and models of the system size dependent assembly of quasi-two dimensional (2D) colloidal crystals within interfacial quadrupole electrodes. Perturbation theory (PT) and Monte Carlo (MC) simulations are used to make thermodynamic predictions of the voltages required to obtain all

~10 Hours

Interactions with the Researcher

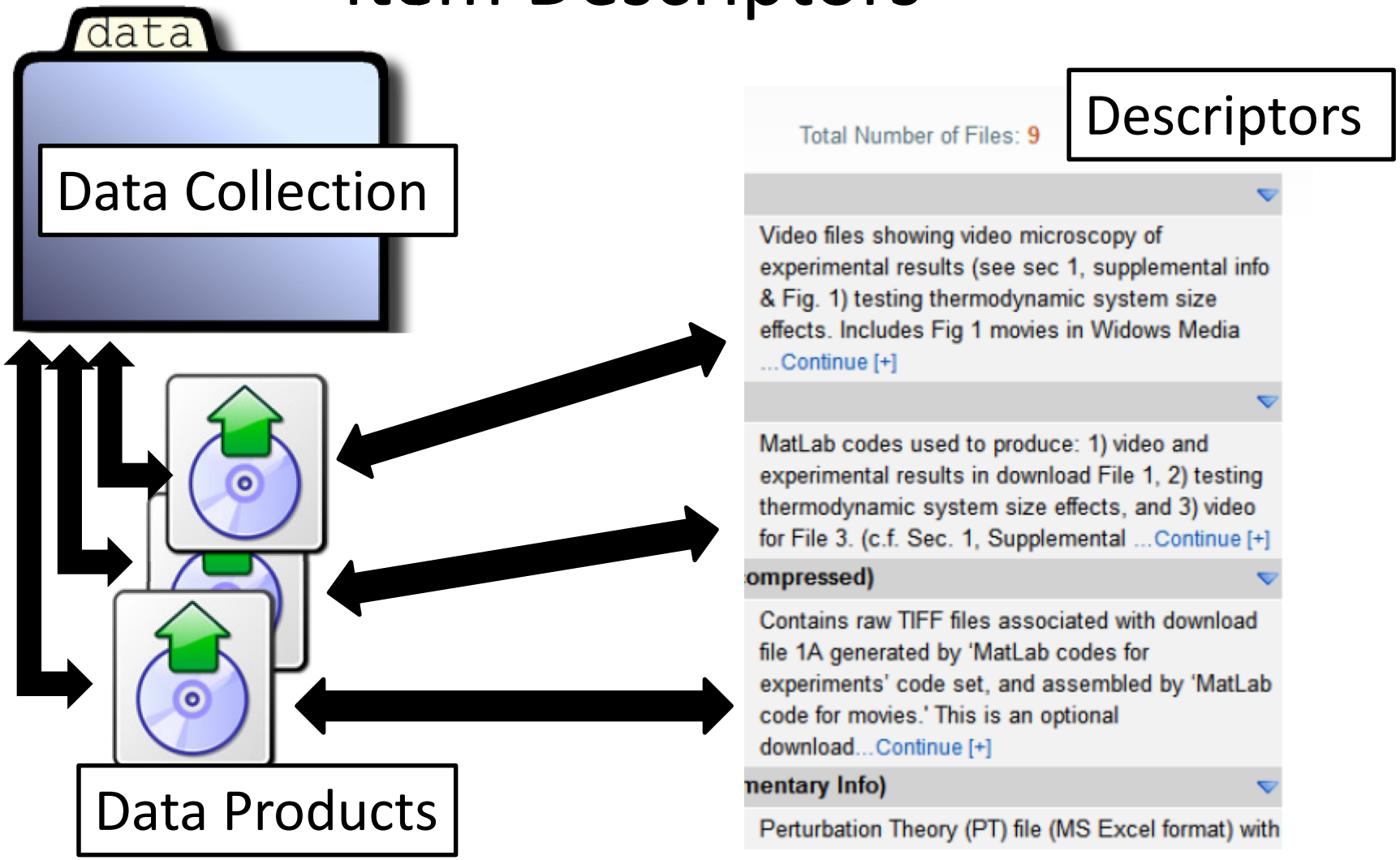
Data Workflow Diagram

~16 Hours



Creating/Modifying Data Item Descriptors

~10 Hours



Format Migration and Annotation

~9 Hours

SigmaPlot™
Exact Graphs and Data Analysis

Migrate

198p-t, s	198p-psi6
0	0.080992031
0.218	0.080105836
0.297	0.080569455
0.406	0.098895949

Annotate

Subplot C	
x-coord	red triangles
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0.297	0.080569455

http://www.sigmaplot.com/products/sigmaplot/sigmaplot-details.php

Uploading Collection

~7 Hours

DATA ASSOCIATED WITH: SIZE DEPENDENT THERMODYNAMICS AND KINETICS IN ELECTRIC FIELD MEDIATED COLLOIDAL CRYSTAL ASSEMBLY.

hdl:TEST/10018

Version: 1 – Released: Mon May 19 11:46:42 EDT 2014

Cataloging Information | **DATA & ANALYSIS** | Comments (0) | Versions

Use the check boxes next to the file name to download multiple files. Data files will be downloaded in their default format. You can also download all the files in a category by checking the box next to the category name. You will be prompted to save a single archive file. Study files that have restricted access will not be downloaded.

Select all files | Download Selected Files | Total Number of Files: 9 | Total Downloads: 1

1_Thermodynamic_sys_size_microscopy_experiments.zip

1_Thermodynamic_sys_size_microscopy_experiments.zip
Zip Archive - 286 MB - 0 downloads
MD5 Checksum: bb3a6d16d2e47b86e73f070e0a4752c2
Download
Video files showing video microscopy of experimental results (see sec 1, supplemental info & Fig. 1) testing thermodynamic system size effects. Includes Fig 1 movies in Widows Media ...Continue [+]

1.1_MatLab code for microscopy experiments data output and movies

1A_MatLab_code_for_experiments_dataoutput_movies.zip
Zip Archive - 582 MB - 0 downloads
MD5 Checksum: f0006261f45d6ee527590f9ec574519e
Download
MatLab codes used to produce: 1) video and experimental results in download File 1, 2) testing thermodynamic system size effects, and 3) video for File 3. (c.f. Sec. 1, Supplemental ...Continue [+]

1.2_TIFF files created by MatLab codes of file 1A (NOTE - contains 25GB original files when uncompressed)

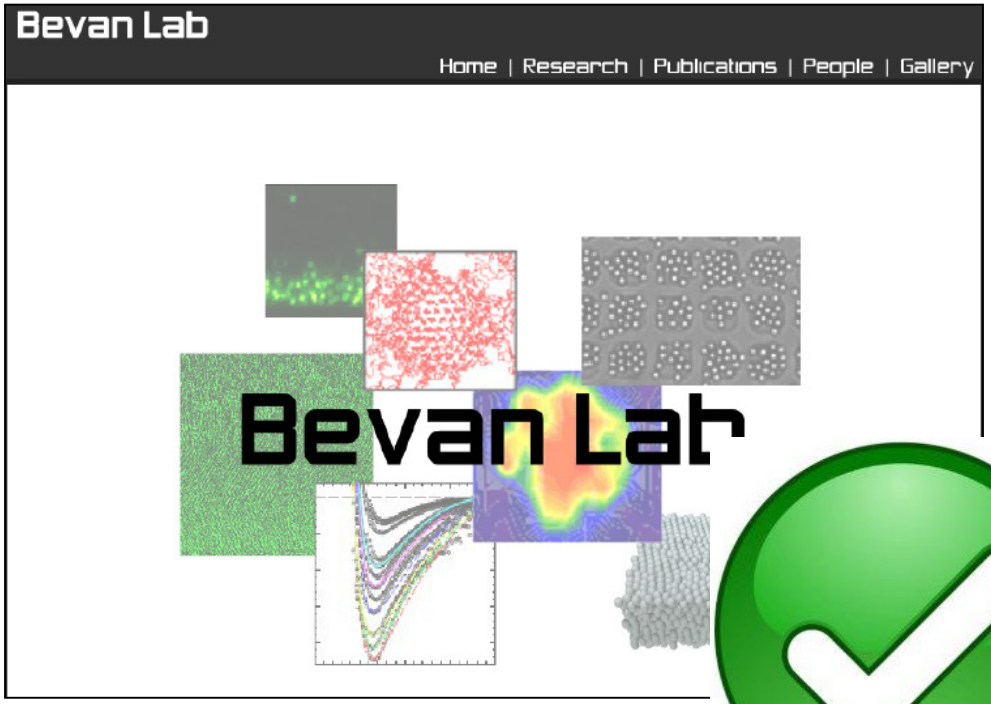
1B_TIFF_output_for_1A_2.zip
Plain Text - 9.7 GB - 1 download
MD5 Checksum: 1f5a84a97ff9ff783db21c7492941238
Download
Contains raw TIFF files associated with download file 1A generated by 'MatLab codes for experiments' code set, and assembled by 'MatLab code for movies.' This is an optional

Which actions to focus on?

~75 hours a collection is not scalable, how much can we reduce hours for those actions?

~~~75 hours~~ -> ~28 hours

~~~



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~6 Hours

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Otherwise we're fumbling in the dark
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~~≈10 Hours~~

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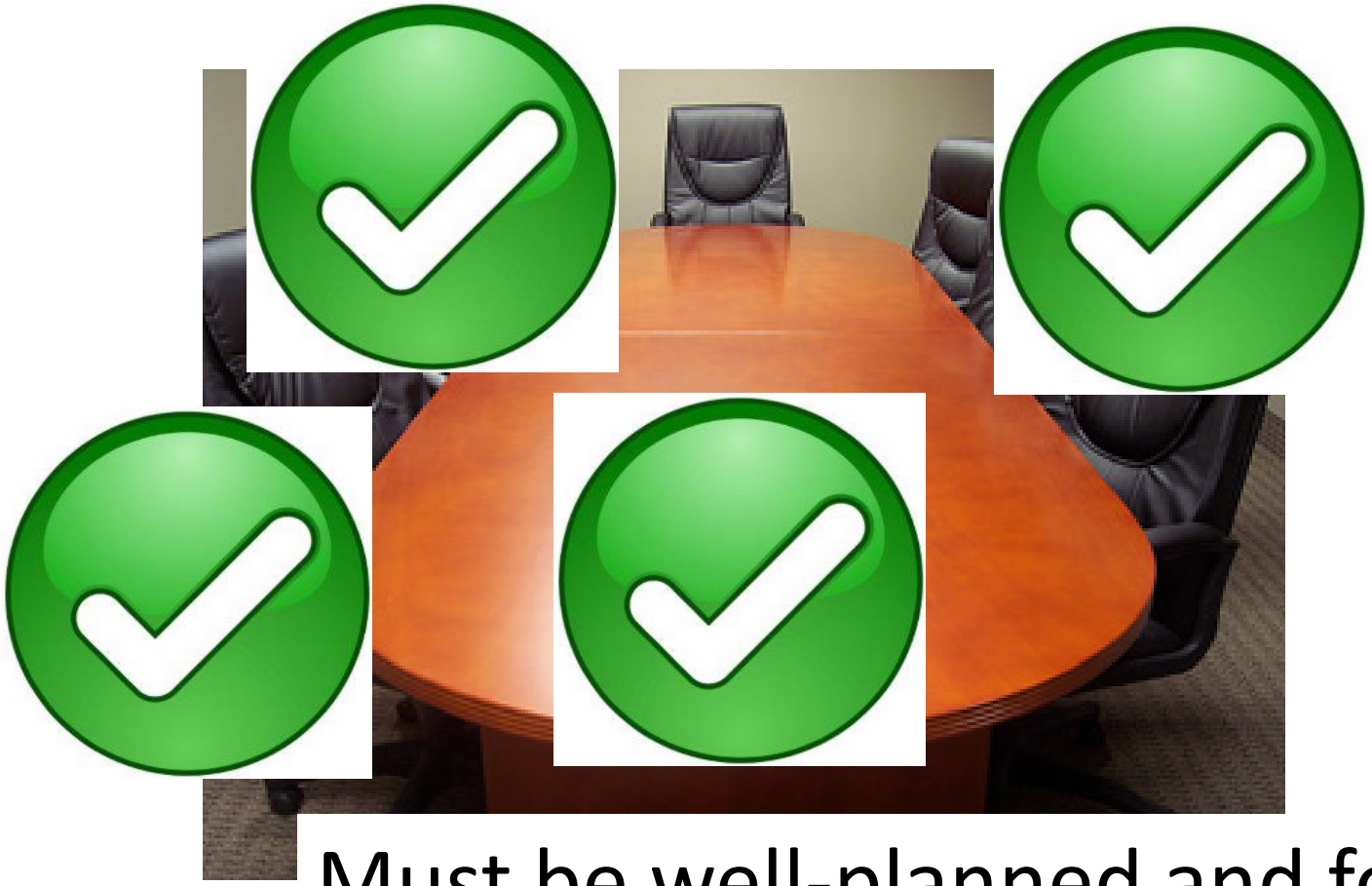
Interactions with the Researcher

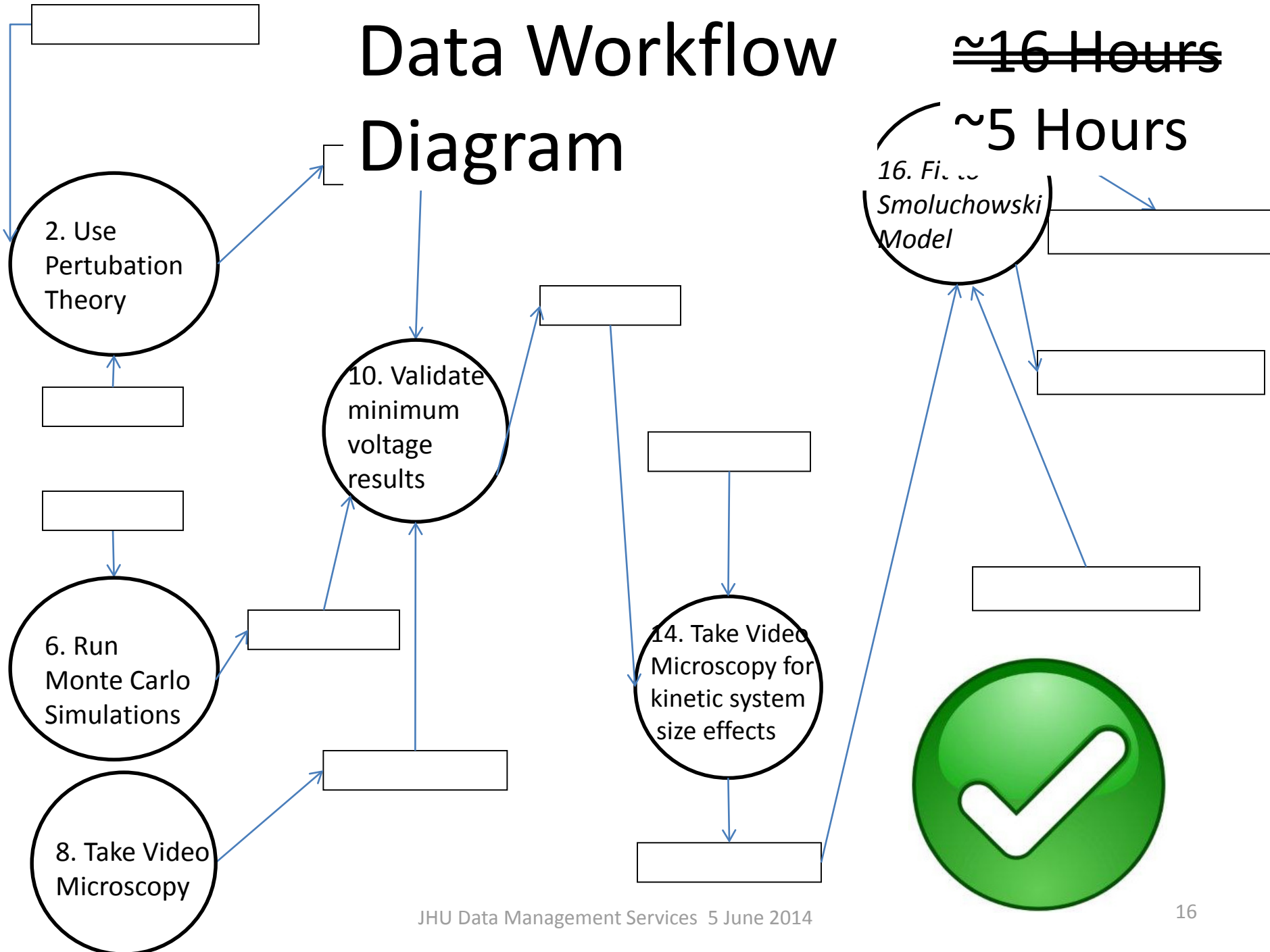


Must be well-planned and focused

Data Workflow Diagram

~~≈16 Hours~~

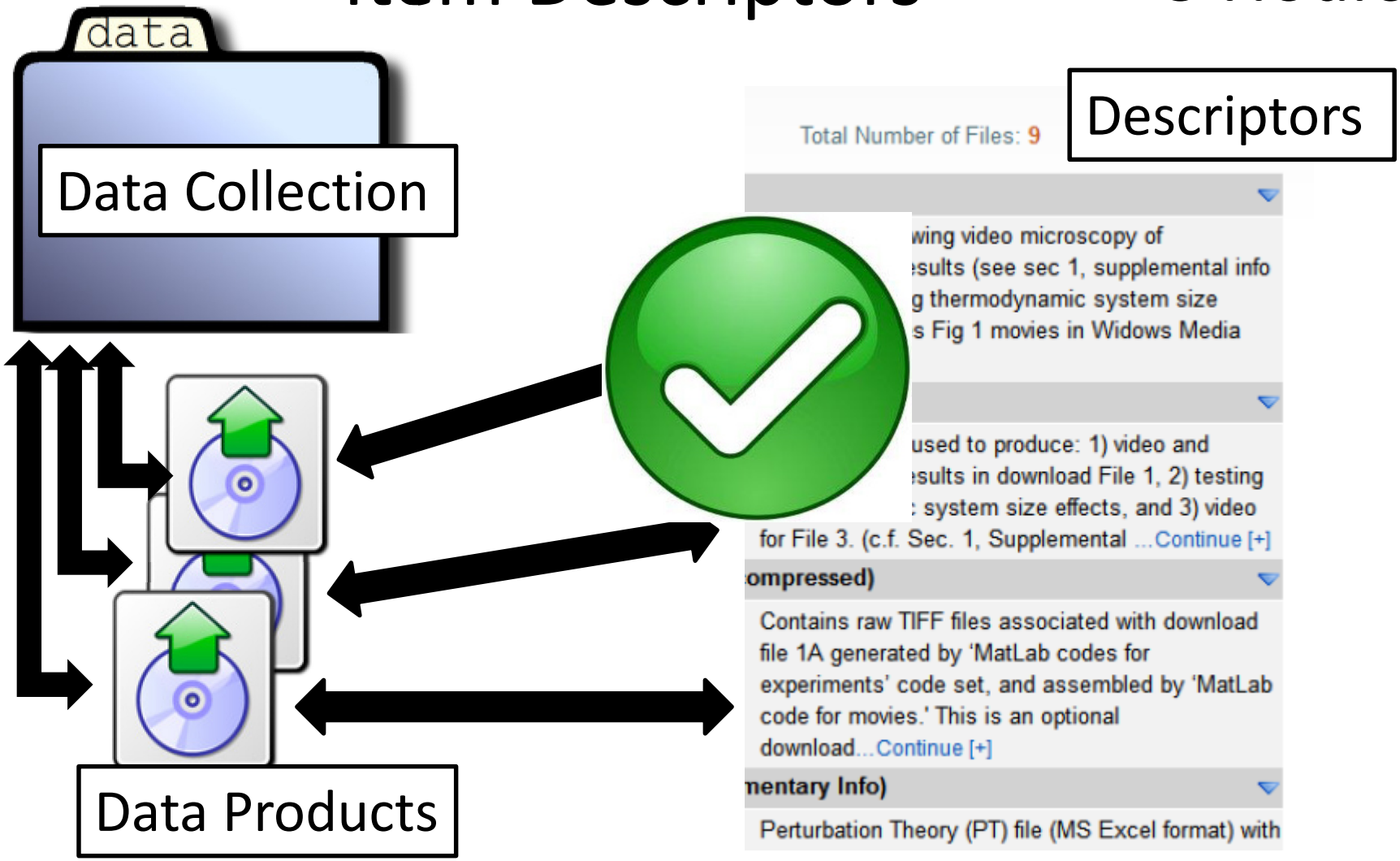
~5 Hours



Creating/Modifying Data Item Descriptors

~~≈10 Hours~~

~3 Hours



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Thanks!

JHU Data Management Services
http://dmp.data.jhu.edu/
datamanagement@jhu.edu

JHU Data Archive - http://archive.data.jhu.edu

Jonathan Petters - jpetter1@jhu.edu